



Fig. 4: Accelerometer

to the controlling unit through some opto-isolator circuitry.

Controlling unit. The controlling unit can be implemented with any microcontroller that has enough memory for your program and inputs/outputs (I/Os) to interface motor drivers and sensing section. You might also like to add an LCD and a keypad. To interface these, you would need additional I/Os.

An additional feature of this robot can be manual control through RF remote control. With this feature, you can sit relaxed on your sofa and direct the cleaner to specific locations using the remote control. In this case also, you will need to interface the RF receiver module to the controlling unit.

Sensing unit. The basic sensing unit can be implemented by using an IR transmitter-receiver pair. The sensors help the robot to detect obstacle in its path. The infrared (IR) LED emits light wavelengths in the IR spectrum, which are invisible to the human eye. The IR receiver can detect these rays. Four such modules can be placed on left, right, front and back of the robot to detect obstacles from all sides and avoid collision.

To make the robot really intelligent, you can use an accelerometer

in the sensing unit. The robot can be programmed to sense the direction of collision with an obstacle using this on-board accelerometer. If head-on collision is detected by the accelerometer, the robot should move backward and change the direction. If an obstacle is hit at an off-angle, the robot should turn away from the direction of the impact. Otherwise, the robot should move in a straight line until it detects a collision.

The vacuum cleaner robot's movement is based upon a random walk around a room, which theoretically will cover the entire area of a room given enough time. The robot is programmed to drive straight until an obstacle is hit. At that point, it will turn and continue driving straight until another obstacle is hit, and so on.

Software. The software decides the intelligence of the overall system as all your logics are implemented here. First, implement the basic algorithm, then keep fine-tuning it to increase the intelligence level. The basic algorithm can make the robot move in a straight line until it detects a collision. Once a collision is detected, the robot will change its direction and again move in a straight line until the next collision.

In order to turn the robot after a collision, one of the wheels is set to reverse while the other wheel keeps moving forward. Take care that the rotating wheel stops spinning while turning. Otherwise, it will interfere with the spin of the robot. The angle of spin can be precisely calculated and controlled as stepper motors are used. Use a wide angle of spin after collision so as to avoid another collision from the same object. The DC motor of the brush can be controlled using PWM technique.

Q2. Please tell me some good books that can help me learn about microcontrollers.

Vinay Kumar R.

Through e-mail

A2. The selection of a perfect book for you will depend on several things like which microcontroller you want to learn (8051, PIC, AVR, etc), your expertise level and programming language preferred (Assembly or 'C'). If you are a beginner, you should start with MSC-51. There are several books available, but for beginners I would recommend 'The 8051 Microcontroller and Embedded Systems' by Muhammad Ali Mazidi and Janice Gillespie Mazidi and 'The 8051 Microcontroller' by Kenneth Ayala. For more insight, you may also consider 'The Final Word on 8051' by Matthew Chapman and 'Programming and Customising 8051 Microcontroller' by Myke Predko.

Answers compiled by EFY technical editor, Ankit Gupta. Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (E-mail: editsec@efyindia.com) and should include name and address of the sender

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